

# Siddharth Saminathan

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## PROFILE

Backend-focused AI engineer with 4+ years shipping production AI systems end-to-end, from data pipelines to user-facing features. Proven ability to own product outcomes: built an AI companion from concept to 450+ users with 13-15% weekly retention, reduced inference cost per conversation from \$1.20 to \$0.023, and cut response latency from 12.5s to 8s. Experienced with LLM APIs, multi-agent orchestration, and real-time personalization, ready to apply this to e-commerce clienteling and conversational BI.

## COMPETENCIES

- AI & Personalization
- Backend & Systems
- Data & Business Intelligence
- Infra & Deployment

## EXPERIENCE

AI Engineer

Omnex Systems (AquaPro AI), Chennai, India  
2025 - Present

- Shipped a production AI system that powers AI-assisted quality and control workflows (DFMEA, PFMEA, Control Plans, Process Flows, 5W2H, 5 Whys, 8D), improving consistency and reducing manual analysis effort for engineering teams, demonstrating ability to build AI-native features for domain-specific business intelligence.
- Designed an orchestration layer using tool-calling: selects tools, routes data across internal services, and executes multi-step workflows end-to-end (generate, validate, update), a pattern directly applicable to building conversational BI layers that replace dashboards.
- Enabled the system to understand production hierarchies and categories, map inputs to the right structures, and suggest updates based on existing data, analogous to personalization intelligence that maps customer data to relevant actions.
- Implemented feedback-aware flows: retrieves similar past concerns, links them, updates records, and suggests impactful actions to users, a core capability for clienteling tools that empower store teams with actionable insights.
- Reduced response latency from ~12.5s to ~8s by restructuring the pipeline and consolidating model calls, ensuring real-time responsiveness for user-facing AI features.
- Improved output consistency and reduced failures with validation layers, retry handling, and fallback logic, critical for reliable LLM API integration in production.
- Implemented observability (structured logging, tracing, and metrics) to monitor system behavior and debug production issues, ensuring operational excellence for AI-native products.

## Co-Founder, AI Product Engineer

Emote (emotenow.app)

2024 - Present

- Built a production AI companion system for hyper-personalized emotional support, users feel remembered and do not start over each session.
- Designed a memory system that tracks emotional context, relationships, and cause-effect patterns across 100s of conversations, improving relevance per user.
- Built the backend from scratch (FastAPI, Redis, PostgreSQL, hosted on Fly.io) with SSE streaming and WebSocket-based real-time pipelines, caching, and multi-model routing.
- Scaled from 0 to 450+ users, with 40+ power users and ~13-15% weekly cohort retention (pre-PMF).
- Reduced inference cost per conversation from ~\$1.20 to ~\$0.023 through model selection, caching, and prompt optimization.
- Improved latency from ~15s to ~3s through pipeline optimization and streaming.
- Built an internal AI product manager that monitors system behaviour, detects anomalies, and improves observability across the production AI stack.
- Built a Reddit bot to automate user acquisition; reduced outreach time from ~6-8 hours to ~30-40 minutes, enabling 50-100 hyper-personalized DMs per session, increasing activation from ~20% to ~75% and weekly retention from ~5% to ~13%.

## Data Engineer

Akkodis AB (Ericsson Client), Sweden

Nov 2022 - Nov 2024

- Owned the end-to-end migration of an enterprise data platform from Azure to on-prem PostgreSQL, ensuring data reliability and consistency through rigorous quality audits, directly enabling business intelligence for cross-functional stakeholders.
- Architected and shipped the full ETL lifecycle for Power BI and Tableau dashboards, replacing manual reporting with automated, data-driven insights used by business owners.
- Redesigned data pipelines powering reporting systems, collaborating with data owners and stakeholders to align data architecture with evolving business needs.
- Drove data quality initiatives during large-scale migration, establishing consistency checks that improved reporting reliability across the organization.

## Data Analyst

Positive Integers Pvt Ltd, Chennai, India

Sep 2019 - May 2020

- Built SQL-based ETL workflows to extract, transform, and load data from operational source systems into reporting layers, enabling downstream business intelligence.
- Built Tableau dashboards for an NBFC to analyze credit risk and loan repayment behaviour, delivering actionable insights for decision-makers.
- Built real-time Grafana dashboards for ID by SPI Cinemas to track full order-to-table lifecycle, supporting operational efficiency. ---

## KEY ACHIEVEMENTS

Shipped Emote, an AI-native personalization engine, from zero to 450+ users with 40+ power users and 13-15% weekly retention, all through organic growth. (<https://emotenow.app/>)

Led enterprise data platform migration (Azure to PostgreSQL) at Akkodis for Ericsson, driving improved reporting reliability and delivery outcomes.

EDUCATION

M.Sc. Statistics and Machine Learning, Linköping University

Sweden · 2020-2022

Shipped an end-to-end ML pipeline for colorectal cancer metastasis prediction from high-dimensional RNA sequencing data (~64,000 features, under 50 samples), demonstrating ability to deliver AI features from raw data to actionable insights.

Deployed autoencoders for feature compression and representation learning, directly applicable to building personalization engines that extract latent user preferences.

Applied PCA for dimensionality reduction and feature selection, mirroring techniques used in clienteling intelligence layers to identify high-value customer segments.

Performed gene set enrichment analysis and statistical modelling to identify biologically relevant patterns, showcasing business-outcome thinking and ability to communicate complex findings to non-technical stakeholders.

B.Tech Computer Science & Engineering, SRM University

India · 2016-2020

SKILLS

|                              |                                                                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI & Personalization         | LLM APIs (OpenAI, Claude, Gemini) · RAG pipelines · embeddings · prompt design · multi-model routing · multi-agent orchestration · memory systems · personalization engines |
| Backend & Systems            | FastAPI · Redis (caching, queues) · PostgreSQL · REST APIs · async pipelines · WebSockets · SSE streaming                                                                   |
| Data & Business Intelligence | SQL · Python · ETL pipelines · Power BI · Tableau · data validation · pipeline optimization                                                                                 |
| Infra & Deployment           | Docker · Fly.io · AWS (basics) · logging · observability · performance optimization                                                                                         |